

Figure 2: Safety Audit Report for Subject-0009.

CLINICAL SAFETY AUDIT REPORT

Subject ID: Subject-0009 **Ground Truth:** MCI

Model Prediction: Normal Control (NC)

Confidence: $P = 0.82$ **Uncertainty:** $U = 0.55$

1. Clinical Audit Summary

Verdict: Suspected False Negative (Missed Diagnosis)

Risk Assessment: High Risk of Discrepancy

Rationale: High uncertainty ($U \approx 0.55$) combined with visible atrophy in Temporal Pole Sup L suggests a missed diagnosis, despite the model's binary classification of NC.

2. Evidence Verification

Anatomical Check:

- **Temporal Pole Sup L:** Atrophy detected ($Z = 1.48$).
- **Hippocampus:** Preserved but warrants monitoring.

3. Agent Recommendations

Next Steps: Manual review required. Consider neuropsychological testing to confirm MCI conversion risk. Re-evaluate segmentation for Temporal Pole region.

Agent Interaction: Orchestrator triggered ANOMALY_INVESTIGATION due to moderate uncertainty and anatomical outliers.